

Academic Qualifications

Year	Degree/Certificate	Institute	CPI/%
2025 - Present	M.Tech Aerodynamics	Indian Institute of Technology, Kanpur	8.63/10.00
2024	B.Tech Aerospace Engineering	Amity University, Mumbai	8.78/10.00
2020	XII (Maharashtra Board)	Thomas Baptista College, Vasai	77.54%
2018	X (Maharashtra Board)	St. Anthony's School, Vasai	91.60%

Projects

- Large Eddy Simulation (LES) of Contrail Formation by Modeling Jet–Vortex Interactions and Ice Microphysics | M.Tech Thesis**
Supervisor: Dr. Pradeep Moise | IIT Kanpur (Feb'26 - Present)
 - Established the temporal turbulent-jet baseline in Incompact3D using LES for aircraft contrail modelling, analysing turbulent kinetic energy (TKE) evolution and one dimensional energy spectra to characterise the development of the turbulent flow.
 - Traced and analysed the existing Fortran implementation from input parameters and boundary conditions through turbulent jet initialization and flow evolution; diagnosed discrepancies in the previous validation and iteratively refined the simulation setup to improve agreement with established turbulent jet behaviour.
 - Compared Dynamic Smagorinsky and explicit high-order low-pass filtering approaches within LES, evaluating TKE evolution and energy spectra to assess their treatment of resolved turbulent scales and identify the configuration most suitable for the temporal jet study.
 - Verified the numerical reliability of the temporal jet simulations through grid convergence analysis and assessed turbulent energy transfer using the Kolmogorov $-5/3$ inertial range scaling, establishing the physical and numerical baseline for subsequent flow interaction studies.
 - Extended the temporal-jet study toward jet-vortex interaction, using the jet's maximum-TKE state to determine the physically relevant interaction onset and developing the framework for subsequent Lagrangian particle transport and ice microphysics modelling of contrails.
- Design, Analysis, Fabrication and Testing of Thrust Vector Control Model Rocket | B.Tech Major Project**
Supervisor: Dr. Amol Kamble | Amity University, Mumbai (Feb'24 - Jun'24)
 - Led a four-member team in developing a Thrust vector controlled (TVC) model rocket for controlled ascent and retro thrust assisted landing, coordinating mechanical design, avionics integration, fabrication and system level assembly.
 - Designed and iteratively refined the TVC mechanism and landing leg assembly using Fusion 360 and SOLIDWORKS, integrating dual servo actuation for multi axis thrust vectoring and optimising the mechanisms for fabrication and reliable assembly.
 - Developed and tuned PID-based thrust-vector control logic in MATLAB/Simulink, evaluating control response and thrust-vectoring requirements for trajectory correction and autonomous landing manoeuvres.
 - Simulated and optimised propulsion and trajectory performance using RockSim and MATLAB/Simulink, iteratively evaluating propulsion and control configurations to achieve the required landing flight performance under an approximately 50% lower impulse constraint than the conventional baseline.
 - Fabricated and integrated the final prototype, 3D printing the TVC and landing assemblies and completing mechanical integration and system level assembly to translate the simulated design into a functional physical model.
- India's Emerging Aerospace Ecosystem: Global OEM Partnerships and Supply Chain Localisation | B.Tech Minor Project**
Supervisor: Dr. Amol Kamble | Amity University, Mumbai (Aug'23 - Dec'23)
 - Investigated India's integration into the global aerospace manufacturing supply chain by examining how partnerships with major aerospace OEMs could build domestic manufacturing and engineering capabilities.
 - Developed a five-level localisation framework spanning assembly, component manufacturing, complex aerostructures, engineering design and indigenous IP ownership; applied it to GE-HAL, Airbus-Tata, and Boeing-Tata partnerships using publicly available 2023 evidence.
 - Assessed manufacturing maturity across the three partnerships, identifying demonstrated complex aerostructure capability, developing assembly/component capability and proposed advanced propulsion manufacturing pathways, alongside supply-chain and sustainability constraints.

- **GPU-Accelerated Finite Volume Method Solver | Course Project**

Course: High Performance Computing with Applications | IIT Kanpur

(Jan'26 - Apr'26)

- **Developed a GPU accelerated 2D compressible-flow solver** using the **Finite Volume Method (FVM)**, implementing the numerical workflow in **CuPy** and **CUDA C** for parallel execution on an **NVIDIA GPU**.
- **Implemented the FVM numerical workflow using second-order MUSCL reconstruction, a Minmod TVD limiter and HLL flux evaluation**, combined with **CFL-based adaptive time stepping** for stable and accurate simulation of compressible flows.
- **Implemented and optimised both Euler and third order Runge Kutta (RK3) time integration**, going beyond the single-scheme requirement, and benchmarked both across increasing grid sizes up to **33.5M cells**, analysing **GPU scaling, computational cost and performance saturation** while maintaining **accurate shock and discontinuity resolution**.
- **Validated the solver against the 2D Sod shock-tube problem and analytical solution**, verifying accurate resolution of **shock waves, contact discontinuities and expansion regions** before large-scale performance benchmarking.
- **Enhanced GPU execution through shared-memory tiling, float4 vectorization and coalesced memory access**, improving data reuse and reducing **global-memory traffic and kernel-launch overhead** during large-scale simulations.
- **Optimised GPU execution through kernel fusion, data reuse and memory-access restructuring**, reducing unnecessary global memory transfers and intermediate data movement while balancing **register usage, memory utilisation and GPU occupancy** through **iterative optimisation cycles**.
- **Profiled the solver using NVIDIA Nsight Compute** to analyse **SM throughput, active warps, L1/DRAM throughput and FP32 instruction throughput**, using hardware-level measurements to identify memory-access and resource-utilisation bottlenecks and guide successive optimisations.
- **Achieved $\sim 30\%$ of peak FP32 GPU throughput across both Euler and RK3 implementations**, exceeding the prescribed **10% target by $\sim 3\times$** on an **NVIDIA RTX 6000 Pro**, with **SM throughput reaching $\sim 78\%$ for Euler and $\sim 69\%$ for RK3** at large grid sizes.

- **2D Backward-Facing Step Flow | Course Project**

Course: Computational Fluid Dynamics and Heat Transfer | IIT Kanpur

(Jan'26 - Apr'26)

- **Investigated flow separation and downstream reattachment** in a 2D incompressible backward-facing step flow using numerical simulation to characterise the resulting recirculation region.
- **Developed a 2D incompressible CFD solver** using the **stream function vorticity formulation** and **second order finite difference discretisation** for the governing Navier Stokes equations.
- **Implemented the numerical solution procedure** for the coupled Stream function, Poisson and vorticity transport equations and resolved the flow field at **Re = 50** on a 300×30 computational grid.
- **Analysed streamlines, velocity and vorticity fields** to identify **flow separation, recirculation and downstream reattachment**, characterising the development of the separated wake behind the step.
- **Verified the numerical formulation through second order truncation error analysis and cell Peclet number assessment**, establishing the accuracy and stability of the finite-difference solution.

- **2D Flow Past a Confined Square Cylinder | Course Project**

Course: Computational Fluid Dynamics and Heat Transfer | IIT Kanpur

(Jan'26 - Apr'26)

- **Investigated the effect of Reynolds number on wake development** behind a confined square cylinder, studying the transition from steady to unsteady flow behaviour.
- **Implemented cylinder and obstacle boundary conditions** in a **Fortran based MAC finite difference CFD solver**, adapting the existing framework for 2D incompressible flow past the cylinder.
- **Configured and executed simulations** on a **non uniform staggered grid** at **Re = 20, 30 and 75**, resolving velocity, pressure and vorticity fields around the cylinder.
- **Characterised the evolution of wake dynamics with increasing Reynolds number**, identifying a **steady symmetric wake at Re = 20**, near wake recirculation and reverse flow at **Re = 30**, and periodic **Kármán vortex shedding at Re = 75**.
- **Analysed centreline and transverse velocity profiles** across Reynolds numbers to quantify changes in **wake recovery, reverse flow regions and downstream flow development**.
- **Interpreted velocity, pressure and spanwise vorticity fields** to relate increasing Reynolds number to **stronger shear layers, extended recirculation and unsteady wake formation**.

- **Linear Stability Analysis of Shear Flows Using the Orr–Sommerfeld Equation | Course Project**

Course: Turbulence | IIT Kanpur

(Jan'26 - Apr'26)

- **Developed a MATLAB based linear stability analysis framework** to evaluate the stability of **parallel shear**

- flows using the **Orr–Sommerfeld equation**, computing unstable eigenmodes and neutral stability characteristics.
- **Computed dominant unstable eigenmodes and disturbance eigenfunctions** at $\alpha = 0.20$ and $Re = 650$, analysing wall-normal and streamwise disturbances to quantify the influence of **base-velocity profiles on instability growth**.
- **Generated neutral stability curves in the (α, Re) parameter space**, identifying instability boundaries and characterising the **onset of transition** and differences in disturbance evolution between the two shear-flow velocity profiles.
- **Swept Open-Fan Rotor: First-Principles Aerodynamic Design and Verified CAD** | Self Project
 - **Designed a swept open-fan rotor for high-subsonic operation**, using first-principles aerodynamic design methods to establish the rotor configuration and design requirements.
 - **Applied actuator-disk sizing and Adkins–Liebeck blade-element momentum theory** to generate the spanwise blade geometry and derive a **44.66° swept-tip configuration**.
 - **Built and independently verified the final CAD blade and periodic-sector CFD domain**, with the aerodynamic design passing **7 internal consistency checks**.
- **High-Pressure Turbine Blade: Internal Cooling Architecture and Geometric Verification** | Self Project
 - **Developed a multi-pass internal cooling architecture** for a high-pressure turbine blade for preliminary aerothermal design studies.
 - **Integrated a network of 32 connected jet passages within the blade CAD model** to develop the internal cooling system.
 - **Performed repeatable geometric verification** of passage connectivity, wall thickness and ligament constraints to establish the CAD model’s readiness for downstream analysis.

Work Experience

- **Internship Trainee | L&T Defence** (Jun’23 - Aug’23)
Mumbai, India
 - **Built a VTOL UAV performance database from scratch**, targeting **100 aircraft** and compiling technical specifications across **9 configuration and performance parameters** for comparative analysis.
 - **Exceeded the 100 aircraft target by researching 200+ VTOL UAVs**, screening available specifications and compiling approximately **195 usable records** across **Fixed-Wing, Quadcopter and Helicopter** configurations.
 - **Evaluated 5 key performance parameters** across the dataset, analysing approximately **60 parameter relationships** to derive configuration-specific trends for preliminary design estimation.
 - **Developed a Python-based estimation tool with bidirectional Maximum Take Off Weight(MTOW)/payload estimation** for **3 UAV configurations**, converting the derived trends into user-driven calculations.
 - **Built a MySQL database with SQL based MTOW and range searches**, allowing users to **input any desired MTOW range** and retrieve comparable aircraft with their recorded specifications.
 - **Integrated UAV data collection, SQL retrieval and Python based parameter estimation** into a **single performance assessment tool**, enabling rapid comparison of existing aircraft and estimation of key values for new configurations.
- **Internship Trainee | Hindustan Aeronautics Limited** (Aug’22 - Sep’22)
Nashik, India
 - **Studied the KOPYO radar system of the Mikoyan MiG-21**, analysing its major subsystems including the exciter, receiver, signal processing and weapons-avionics interface.
 - **Analysed the Master Oscillator and Exciter Unit**, understanding radar frequency generation and gaining practical exposure to **snag identification, testing and overhaul** of radar sub-units.
 - **Studied integrated aircraft avionics**, covering **RWR, INS/GPS, IFF and Air Data Systems** and their roles in navigation, threat detection and aircraft operations.
 - **Observed aircraft manufacturing, assembly and overhaul workflows** across radar, fly-by-wire, navigation, armament and final-assembly facilities, gaining exposure to **aircraft-system integration and maintenance practices**.
- **Internship Trainee | Space Technology and Aeronautical Rocketry** (Jun’22 - Jul’22)
Surat, India
 - **Led a six-member team** in designing and developing a **1U CubeSat structure**, meeting the **250 g mass constraint** while accommodating **PCB and deployable antenna integration** requirements.
 - **Designed and refined the CubeSat structure in Autodesk Fusion 360**, optimising panel geometry, screw-based

assembly and **3D printing constraints** for manufacturability.

- **Evaluated structural integrity using FEA across 20–100 N loading conditions**, analysing stress and deformation to refine the structure before fabrication.
- **Reduced the final structural mass to approximately 194 g**, achieving a **23.2% margin below the 250 g limit** while maintaining structural integrity under the evaluated loading conditions.
- **Fabricated and assembled the final structure using 3D printed components**, translating the CAD design into a **physical prototype** for subsystem integration and testing.

Positions of Responsibility

- **Teaching Assistant | AE646: Scientific Machine Learning for Fluid Mechanics | IIT Kanpur** (Aug'26 - Present)
 - **Support students throughout the course**, attending lectures and resolving queries on **machine learning concepts and their applications to fluid mechanics**.
 - Mentored students' teams on Scientific Machine Learning for Fluid Mechanics, providing technical guidance on PINNs, DeepONet, FNOs, CNNs, Autoencoders, and reduced-order modeling providing technical guidance on implementation, debugging, and model evaluation.
 - Evaluated project proposals, code reproducibility, technical presentations, and viva examinations, while coordinating assignments and course activities for a project-based graduate course.
 - **Manage project-based evaluation** by developing an **Excel based assessment and project tracking system**, reviewing reports and presentations and conducting **technical vivas** to assess individual understanding and contributions.
- **PG Student Guide | Centre for Mental Health and Wellbeing (CMHW) | IIT Kanpur** (Jul'26 - Present)
 - **Guided and mentored 6 students**, helping them adapt to the academic and campus environment at **IIT Kanpur** and addressing concerns related to their transition to the institute.
 - **Supported students in navigating academic and institute related matters**, coordinating with relevant mentors and institutional resources when required.
- **Student Relations Head | Society of Aerospace Engineers IIT Kanpur** (Mar'26 - Present)
 - **Coordinate SAE Coffee Hours twice a month**, managing member communication and event execution to facilitate regular student engagement.
 - **Managed Research Scholars Day and Convocation responsibilities** for the Department of Aerospace Engineering, coordinating **chief guest hospitality, volunteer allocation and event execution**, including airport pickup, accommodation and on site requirements.
 - **Coordinated vendors for Convocation**, negotiating prices and finalising arrangements for flower, decoration and cloth requirements; also supported **script writing, proofreading and event coordination** for SAE and institute events.
- **Teaching Assistant | AE211: Incompressible Aerodynamics | IIT Kanpur** ([Jan '26 - Apr '26])
 - **Checked and evaluated assignments for approximately 80 students**, ensuring consistent assessment and timely completion of course evaluation.
 - **Supported invigilation during tests and examinations**, monitoring examination halls, **upholding academic integrity** and ensuring the smooth conduct of assessments.

Technical Skills

- **CFD & Simulation:** Incompact3D, ANSYS Fluent, ANSYS Mechanical, ANSYS Workbench, Finite Element Methods, Turbulence Modelling, Numerical Methods
- **CAD & Engineering Design:** Fusion 360, SOLIDWORKS, Siemens NX, AutoCAD, OpenRocket, RockSim,
- **Post Processing & Data Analysis:** ParaView, Tecplot, CFD Post, NumPy, Pandas, Matplotlib
- **Programming & Computing:** Python, MATLAB, Fortran, C basics, C++ basics, CUDA C, CuPy, High Performance Computing, MySQL, Simulink
- **Engineering & Productivity Tools:** Microsoft Excel, LaTeX
- **Soft Skills:** Problem Solving, Analytical Thinking, Technical Communication, Teamwork, Leadership, Adaptability

Scholastic Achievements

- **Earned a Letter of Recommendation from the Director, STAR** for contributions to 1U CubeSat structural design, FEA and fabrication, with recognition for **problem-solving, critical thinking and teamwork**.

- Earned an **A** grade in **IDC606: High Performance Computing with Applications** at **IIT Kanpur**.

Relevant Courses

**UG Courses*

Mathematics for Aerospace Engineering	Aerodynamics-I
Viscous Flows	Finite Element Methods for Fluid Dynamics
Aerodynamics-II	Turbulence
Computational Fluid Dynamics and Heat Transfer	High Performance Computing with Applications
Aircraft Structures*	Aircraft and Rocket Propulsion*
Flight Mechanics*	Computational Fluid Dynamics*
Computational Structural Analysis*	Business Management (Minor Degree)*

Certifications

- **Siemens NX Mastery: Advanced Design & Applications**, Siemens, Coursera
- **Intro to Siemens NX: Engineering Essentials and Part Design**, Siemens, Coursera
- **Simulation using ANSYS**, UdeMy

Extra-Curricular Activities

- **Led the football team** at the Department of Aerospace Engineering Sports Day, securing the **Gold Medal**.
- Secured a **Gold Medal in Throwball** at the same event.

Languages

- English | Marathi | Hindi